Victor Carmona

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Research Group: Mathematical Structures in Physics
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Research Interests

My research interests include higher categories, higher algebra, homotopy theory, pr(operad)s and variations, functor/manifold calculus, (derived) algebraic/differential geometry and mathematical physics.

Education

Universidad de Sevilla — Oct 2018 – Jul 2023
Ph.D. in Mathematics, Advisors: R. Flores & F. Muro
Department of Algebra

Universidad de Salamanca — Sep 2016 – Jul 2017
Master's Degree in Mathematics and Physics

Universidad de Salamanca — Sep 2013 – Jul 2016
Bachelor's Degree in Mathematics

Universidad Complutense de Madrid — Sep 2012 – Jul 2013
First year of double Bachelor's Degree in Mathematics and Physics

Academic Positions

MPI-MiS Leipzig — Sep 2023 – Current
Postdoctoral fellowship
Mathematical Structures in Physics

Universidad de Sevilla — Oct 2018 – Jul 2023
Ph.D. Student with FPU fellowship
Department of Algebra

Universidad de Sevilla — Jan 2018 – Sep 2018
Research fellow
IMUS

Academic Visits

Universidad Autónoma de Madrid — May 2025
Host: F. Cantero Morán
Departamento de Matemáticas

Institut Montpelliérain Alexander Grothendieck (IMAG) — Apr 2025
Host: D. Calaque
Geometry, Topology and Algebra (GTA) Section

Université Sorbonne Paris Nord — Nov 2024
Laboratoire Analyse, Géométrie et Applications

Technische Universität München — Mar 2024
Host: C. Scheimbauer
Department of Mathematics

Université Toulouse III-Paul Sabatier
Host: J. Bellier-Millès — Jun 2023
Department of Mathematics

University of Nottingham
Host: A. Schenkel — Feb 2023 & Jun 2024
School of Mathematical Sciences

Institut Montpelliérain Alexander Grothendieck (IMAG)
Host: D. Calaque — Mar 2022 – Jul 2022
Geometry, Topology and Algebra (GTA) Section

Universidade de Santiago de Compostela
Hosts: L. Alonso-Tarrio & A. Jeremías-López — Jun 2021 – Jul 2021
Department of Algebra

PUBLICATIONS

C^*-categorical prefactorization algebras for superselection sectors and topological order
w/ M. Benini, P. Naaijkens & A. Schenkel
Accepted in *Communications in Mathematical Physics* (2025)
Available at https://arxiv.org/abs/2505.07960

On the equivalence of AQFTs and prefactorization algebras
w/ M. Benini, A. Grant-Stuart & A. Schenkel
Accepted in *Letters in Mathematical Physics* (2025)
Available at https://arxiv.org/abs/2412.07318

Algebras over not too little discs
w/ D. Calaque
Accepted in *Communications in Mathematical Physics* (2025)
Available at https://arxiv.org/abs/2407.18192

Enveloping operads and applications
Journal of Pure and Applied Algebra, Vol 229, Issue 12 (2025)
Available at https://arxiv.org/abs/2407.18190

Cloning systems and action operads
w/ J. Aramayona, F. Cantero Morán & J.J. Gutiérrez
Accepted in *Publicacions Matemàtiques* (2025)
Available at https://arxiv.org/abs/2303.09873

Strictification theorems for the homotopy time-slice axiom
w/ M. Benini & A. Schenkel
Letters in Mathematical Physics, Vol 113, no. 1 (2023)
Available at https://arxiv.org/abs/2208.04344

New model structures for Algebraic Quantum Field Theory
Letters in Mathematical Physics, Vol 113, no. 2 (2023)
Available at https://arxiv.org/abs/2107.14176

When Bousfield localizations and homotopy idempotent functors meet again
Homology, Homotopy and Applications, Vol 25, no. 2 (2023)
Available at https://arxiv.org/abs/2203.15849

Homology and cohomology of finite spaces
w/ C. Maestro, F. Sancho de Salas & J.F. Torres Sancho
Journal of Pure and Applied Algebra, Vol 224, Issue 4 (2020)
Available at https://arxiv.org/abs/1809.08047

PREPRINTS

Relative operads model ∞-operads
w/ K. Arakawa & F. Pratali
Available at https://arxiv.org/abs/2512.16374

Additivity of constructible factorization algebras over manifolds with corners
w/ A. Svraka
Available at https://arxiv.org/abs/2510.26504

A simple characterization of Quillen adjunctions
Available at https://arxiv.org/abs/2406.02194

A model structure for locally constant factorization algebras
w/ R. Flores & F. Muro
Minor revision in *Algebraic & Geometric Topology* (2025)
Available at https://arxiv.org/abs/2107.14174

INVITED TALKS

Algebras over not too little discs and TQFTs — Dec 2025
Stockholm University

Algebras over not too little discs and TQFTs — Dec 2025
Uppsala University

C^*-categorical prefactorization algebras of superselection sectors — May 2025
Universidad Autónoma de Madrid

Enveloping operads; one construction to rule them all — Mar 2025
Université de Strasbourg

The ∞-equivalence between AQFTs and Factorization Algebras — Feb 2025
Mathematics and Theoretical Physics (MTP) Seminar, University of Hertfordshire

AQFTs vs. Factorization Algebras: toward a higher comparison — Dec 2024
MPIM (Bonn)

AQFTs vs. Factorization Algebras: toward a higher comparison — Nov 2024
Après-midi parisienne de topologie algébrique, Université Paris Cité

Sistemas de clonado, G-opéradas y grupos de Thompson — Oct 2024
XXII Encuentro de Jóvenes Topólogos, Universidad de Salamanca

Formality of the little n-disc operad and knot theory — Jul 2024
European Talbot Workshop, HCM (Bonn)

Additivity and descent of constructible factorization algebras — Jun 2024
AQFTUK, University of York

Todo lo que baja, suma — Jun 2024
MCHAGTAF, Universidad Autónoma de Madrid

Derived Morita Theory Jan 2024
Congreso bianual de la RSME, Universidad Pública de Navarra

Enveloping operads; one construction to govern them all Jun 2023
ALPE, Université Toulouse III-Paul Sabatier

Renormalization of TFTs via Factorization Algebras Feb 2023
Quantum Maths Seminar, University of Nottingham

An elementary example of deformation quantization via factorization algebras Feb 2023
Encuentro de Jóvenes RSME – Topology session, Universidad de León

A title containing the words "localization", "operads" and "quantum field theories" Jan 2023
Operads in Málaga, Universidad de Málaga

Un título que contiene las palabras "localización", "opéradas" y "grupos de Thompson" Jan 2023
Seminar of Algebra, Universidad de Sevilla

Deformando las matemáticas Dec 2022
LMSI, Universidad de Sevilla

Stabilization of operadic algebras and functor calculus Nov 2022
Algebra & Number Theory Seminar, University of Maine

Strictifying the time-slice axiom Jul 2022
AQFTUK, University of York

Renormalization of TFTs and the homotopy time-slice axiom of AQFTs Jun 2022
Séminaire AGATA, IMAG

Derived localization of operads Feb 2022
PG Forum, University of Lancaster

Two instances of homotopy theory in physics: Factorization Algebras & AQFTs Dec 2021
Quantum Maths Seminar, University of Nottingham

Bisturí. Tengo que cortar esta asa Feb 2021
PhD Seminar, Universidad de Sevilla

Un paseo por el mundo de las opéradas; ahora sin tropiezos Nov 2020
I Jornadas Topológicas Virtuales, Online

Homotopía y Variedades: un ejemplo de co-operación Jan 2020
Encuentro de Jóvenes RSME – Topology session, Universitat Jaume I

Dulce introducción al mundo estable Oct 2019
VIII Encuentro de Jóvenes Topólogxs, Universidad de Santiago de Compostela

A tour through the world of operads Jul 2019
Young Topologists Meeting [YTM], EPFL

A model for factorization algebras Jul 2019
MCHAGT, Universidad de Santiago de Compostela

Localization techniques Jul 2018
Young Topologists Meeting [YTM], University of Copenhagen

Localiza y vencerás — Apr 2018
IUFFyM Seminar, Universidad de Salamanca

Spectra and Motives — Mar 2018
PhD Seminar, Universidad de Sevilla – IMUS

Towards Poincare-Verdier duality — Nov 2017
Seminar of Algebra, Universidad de Sevilla

Foundations of jet bundles — Mar – Jun 2016
IUFFyM, Universidad de Salamanca

Conferences (co)organized

Special session in Congreso Bienal de la RSME — Jan 2026
Universidad de Alicante

7th BYMAT Conference: Bringing Young Mathematicians Together — Nov 2025
(Member of the scientific committee)
Universidad de Sevilla

Regular lecture series: The theory and practice of ∞-categories — May - Jul 2025
Max-Planck Institute for Mathematics in the Sciences

Special session in VII Congreso de Jóvenes Investigadores de la RSME — Jan 2025
Universidad del País Vasco [UPV/EHU]

An Allegedly Somewhat Friendly Introduction to Infinity-Operads — Oct 2022
Universidad de Sevilla – IMUS (Hybrid)

XXVI Encuentro de Topología & IX Encuentro de Jóvenes Topólogxs — Oct 2021
Universidad de Sevilla – IMUS

Métodos Categóricos y Homotópicos en Álgebra, Geometría y Topología — Jun 2021
Universidad de Sevilla – IMUS

IMUS Virtual Series: Operads in Topology — Apr – May 2020
Instituto de Matemáticas de la Universidad de Sevilla [IMUS] (Online)

IMUS Virtual Series: Junior Topology Series — Mar – Jul 2020
Instituto de Matemáticas de la Universidad de Sevilla [IMUS] (Online)

PhD Seminar — Sep 2017 – Oct 2018
Instituto de Matemáticas de la Universidad de Sevilla [IMUS]

Courses Attended (selected)

QFT and Topological Phases via Homotopy Theory and Operator Algebras — Jul 2025
Twinned Workshop: MPIM (Bonn) & CMSA (Harvard)

Gray Products and Lax Constructions — Feb 2025
Technical University of Munich – Research Center Raitenhaslach

Operads in deformation theory and homotopical algebra — Jul 2024
European Talbot Workshop

Summer school on h-principles — Jun 2024
ICMAT

Categorical symmetries in Quantum Field Theory Sep 2023
NCCR Swiss-MAP Research Station

Cellular E_k-algebras May 2021
Oberwolfach (Hybrid)

EAST 2019: European Autumn School in Topology Sep 2019
University of Utrecht

SMS: Derived Geometry and Higher Categorical Structures in Geometry and Physics Jun 2018
The Fields Institute

Rigidity and algebraic models in stable homotopy theory Apr 2018
University of Copenhagen

Tensor triangular geometry and equivariant stable homotopy theory Mar 2018
University of Copenhagen

Riemann–Hilbert Correspondences Feb 2018
Università di Padova

Topos theoretic approach to Quantum Physics Oct 2018
Basque Center for Applied Mathematics [BCAM]

Introduction to Geometric Analysis: The Atiyah-Singer Index Theorem Jun 2017
Basque Center for Applied Mathematics [BCAM]

Teaching Experience

Max-Planck Institute for Mathematics in the Sciences
Academic year 2024 – 2025
The theory and practice of ∞-categories. 20 hours

Universidad de Sevilla
Academic year 2021 – 2022
Linear Algebra. 60 hours (BSc in Mathematics)
Academic year 2020 – 2021
Linear Algebra. 40 hours (BSc in Statistics)
Algebraic Structures and Galois Theory. 20 hours (BSc in Mathematics)
Academic year 2019 – 2020
Linear Algebra. 30 hours (BSc in Statistics)
Algebraic Structures and Galois Theory. 30 hours (BSc in Mathematics)

Reference Letters

Javier Aramayona (*CSIC – ICMAT*, javier.aramayona@icmat.es)
Collaborator

Damien Calaque (*Université de Montpellier – IMAG*, damien.calaque@umontpellier.fr)
Collaborator

Ramón Flores (*Universidad de Sevilla*, ramonjflores@us.es)
PhD advisor

Fernando Muro (*Universidad de Sevilla*, fmuro@us.es)
PhD advisor

Alexander Schenkel (*University of Nottingham*, alexander.schenkel@nottingham.ac.uk)
Collaborator